Correlated residuals in a three-trait QTL SEM: identification, a replication of Mi et al. (2010), and a bow-free estimator

27 October 2026

Abstract

Mi, Eskridge, Wang, Baenziger, Campbell, Gill, Dweikat, and Bovaird asked, in the discussion of their 2010 QTL structural-equation paper, whether trait residuals can be correlated after environment and block effects have been removed. On the three-trait pattern printed in their equations (7)–(8), a diagonal residual matrix is globally identified, each single residual covariance on that pattern collides, and each of the three bow-free patterns is a left inverse of the covariance. Those statements are Lean proofs. This paper then runs three seeded simulations. A first reading of equation (7), with one minus in front of M, does not reproduce Table 2's total effect 0.75 or the single-trait power 97% for the third QTL on the first trait. The minus signs sit on individual entries. That glyph reading reproduces the parameter block, including 0.75, and lifts the single-trait power from 13% to 86% in 1000 replicates. The multi-trait powers 77 and 96, and the last eleven points of that single-trait power, remain. A bow-free Monte Carlo shows that the diagonal plug-in invents a direct edge and biases two QTL coefficients, while `recoverBow` covers the path and QTL coefficients at the delta-method rate. A third run replaces the path coefficients by a fit to three phenotypic correlations from Campbell et al. (2003). It is a realism check on raw correlations, not a replication and not a fit to the wheat sample.

Question, source, and answer

Question	Can the trait residuals in the Mi et al. (2010) QTL SEM be correlated after environment and block effects are removed? Printed page 18 (PDF page 20).
Source	Mi et al., SAGMB 9 (2010), article 38. The request is not a numbered conjecture.
Answer on the printed pattern	On equation (7), diagonal residuals are a left inverse of the covariance. Each single residual covariance collides. Each bow-free missing-edge pattern is a left inverse, and the plug-in is almost surely consistent. Theorems 246, 114, 187, and 270.
What the simulations add	A seeded replication, the bias and size distortion of the diagonal fit, and coverage of `recoverBow`. These are sampled experiments, not Lean proofs.
Still open	A fit to the wheat measurements, the general mixed-graph characterization, and a Lean central limit theorem.

The manuscript date 27 October 2026 is an internal date. It is not an external publication date. Elapsed times below are measured to that date. A year-only source is reported as a range.

Source	Earliest date used	Inspected version	Elapsed to 27 Oct 2026
Brito and Pearl, bow-free identification	2002 (year)	SEM 9, 459–474	23–24 years
Campbell et al., chromosome 3A QTLs	2003 (year)	Crop Sci. 43, 1493–1505	22–23 years
Li et al., structural QTL model	21 July 2006	PLoS Genet. 2, e114	20 years, 3 months, 6 days
Mi et al., the source question	2010 (year)	SAGMB 9, article 38	15 years 10 months to 16 years 10 months
Valente et al., recursive search	June 2010 (month)	Genetics 185, 633–644	16 years, 4 months (month precision)
Drton, Foygel, and Sullivant	arXiv 4 Mar 2010; online 13 May 2011	Ann. Statist. 39, 865–886	15 years, 5 months, 14 days from the journal date
Foygel, Draisma, and Drton, half-trek	arXiv 27 July 2011; online 3 Oct 2012	Ann. Statist. 40, 1682–1713	14 years, 24 days from the journal date
Hidalgo-Contreras et al.	28 September 2021	Agronomy 11, 1953	5 years, 29 days
This manuscript	27 October 2026	internal manuscript date	—

1 The question, as they pose it

The source is Mi et al., Statistical Applications in Genetics and Molecular Biology 9 (2010), article 38. On printed page 18 (PDF page 20) they write:

> Here, we assumed that the residual errors of the traits were independent which we believe is a reasonable assumption since we removed the effects of environment by using, for the y values in equation (2), the residuals from a main effects linear model with environment and block within environment as the main effects. However, even after removing environment main effects, the traits may still be correlated and the trait error terms can be modeled as being correlated with each other.

(taken from a cited source) The parent request is that extension. It is not a numbered conjecture. Mi's 2009 dissertation (ProQuest AAI3350451) is the obvious place a fuller design might have been written. It is restricted to a ProQuest preview. It was not consulted.

2 The restriction

Equation (7) is the only fully printed structural pattern. On PDF page 10 the equals sign ends at $x = 145.9$ points and the parenthesis of M begins at $x = 148.1$. No operator fits between them, so the display is $y = My + Gx + \varepsilon$. The minus at $x = 189.1$ is inside M, on the middle row, and three further minuses sit on G_{11}, G_{22}, and G_{32}. *(our reading of the glyph positions)*

$$M = \begin{pmatrix} 0 & 1/2 & 1/4 \\ 0 & 0 & -1/2 \\ 0 & 0 & 0 \end{pmatrix}, \qquad G = \begin{pmatrix} -1/8 & 1/4 & 1/2 \\ 1/2 & -1/2 & 1/2 \\ 1/4 & -1/8 & 1/4 \end{pmatrix}.$$

Equation (8) is $\Psi = \mathrm{diag}(1.6, 1.8, 2.5)$. *(taken from a cited source)* In $y = By + Gx + \varepsilon$ one then has $B = M$. A first reading placed the interior minus in front of M and took every entry of G as

positive. That reading is the population of the first replication below. The zero pattern is the same either way: the free positions are still $(1,2)$, $(1,3)$, and $(2,3)$. The wheat Figure 5 graph has four traits and is not this matrix. The restriction is covariance identification for this three-trait recursive pattern: diagonal Ψ, one extra residual covariance, and the three ways to drop one edge and free the residual pair that edge had used. Singular Ψ is outside the claim. The general mixed-graph characterization is not this restriction.

3 What we proved

The following are Lean proofs on the positive-definite domain. The axioms are `propext`, `Classical.choice`, and `Quot.sound`. There is no `sorry`. *(our own Lean proof)*

- `recover_sigma`: diagonal residuals and positive-definite Φ are a left inverse of the joint covariance on the equation-(7) support.
- `bow12_same_law` and its two siblings: each one-edge residual extension collides, for every exogenous Γ.
- `recover_bow`, `recover_bow01`, `recover_bow23`: each bow-free missing-edge pattern is a left inverse.
- `unitUpper_factor_unique` and `sigmaRec_injective`: in every trait and covariate dimension, a recursive pattern with positive diagonal residuals has a unique unitriangular factorization, and the joint blocks determine (B, Ψ, Γ, Φ).
- `plug_in_consistent_ae` and the three bow-free analogues `plug_in_bow_consistent_ae`, `plug_in_bow01_consistent_ae`, `plug_in_bow23_consistent_ae`: pairwise i.i.d. samples with integrable coordinate products make those plug-ins almost surely consistent.

The Gaussian delta-method formula for the diagonal plug-in is a written derivative together with Isserlis' covariance, checked numerically in `delta_method.py`. *(our own written proof, and our own simulation)* It is not a Lean central limit theorem, and it is not listed among the proved results.

Statement	Lean declaration	File and lines
Diagonal left inverse	`recover_sigma`	`Diagonal.lean`, 246–279
One-edge collision	`bow12_same_law` and two siblings	`Collision.lean`, 114–160
Bow-free left inverse	`recover_bow`	`BowFree.lean`, 187–213
Other missing edges	`recover_bow01`, `recover_bow23`	`BowMissing.lean`, 148 and 353
Triangular factor	`unitUpper_factor_unique`	`Triangular.lean`, 68–135
Recursive identification	`sigmaRec_injective`	`Triangular.lean`, 298–317
Almost-sure consistency	`plug_in_consistent_ae` and `plug_in_bow_consistent_ae`	`Law.lean`, 174–199; `BowConsistency.lean`, 270–304

The links are pinned to commit `05d69ea141adf3d5ee936d3b48061bde241250dc`, which contains these Lean files. The simulation script and the JSON outputs are commit `5cd3c9f6d2f6e38f25e32146d74d55f3778436e1`. Appendix A prints the central proof bodies.

Brito and Pearl (Structural Equation Modeling, 2002) prove that a recursive linear model is identifiable when no residual covariance joins a variable to its direct effect. Our three bow-free left inverses are the three-trait case of that pattern, proved from the printed matrices rather than cited as a substitute for the algebra. Drton, Foygel, and Sullivant (Annals of Statistics, 2011)

give a global criterion for mixed graphs; the half-trek criterion of Foygel, Draisma, and Drton (Annals of Statistics, 2012) is a generic sufficient condition. Neither is re-proved here, and the 2011 characterization stays a note with no inventory id. Li et al. (PLoS Genetics, 2006) and Valente, Rosa, de los Campos, Gianola, and Silva (Genetics, 2010) are earlier structural and recursive treatments of multiple traits. Hidalgo-Contreras, Salinas-Ruiz, Eskridge, and Baenziger (Agronomy, 2021) fit a later Nebraska wheat SEM; their lines are a 2013 F6 panel, not the 1999–2001 RICLs, and none of their estimates enter the simulations.

4 Replication design

The printed design, PDF pages 9–10, is 100 replicates of 150 RIL lines, one chromosome of 100 cM, 11 evenly spaced markers, additive QTLs at 22, 42, and 78 cM, phenotypes from equations (7)–(8), and a test every 1 cM. QTL genotypes are coded $+1$ and -1. Cofactors are all markers except the flanking markers. The printed critical values are 14.86 and 10.60. *(taken from a cited source)* Those two numbers are the chi-square 0.005 quantiles on 4 and 2 degrees of freedom (14.860 and 10.597). The paragraph on PDF page 9 writes the Bonferroni approximation $\chi^2_{\alpha/M,\,m+1}$. Ten marker intervals and $\alpha = 0.05$ give the tail probability 0.005, with $m + 1 = 4$ for three traits and $m + 1 = 2$ for one trait. *(our own calculation, from the cited formula)*

The paper does not state the following, so each is our assumption. *(our assumption)*

1. The map function is Haldane, with no interference.
2. An RIL is an F1 selfed for 12 generations. A leftover heterozygote is coded ± 1 with equal probability. There is no segregation distortion.
3. The eleven markers are at $0, 10, \ldots, 100$ cM.
4. The analysis uses the centered covariance with divisor n, which is the fitting function (6). We did not run SAS PROC TCALIS.
5. The flanking expectation is the conditional mean of the QTL genotype under that same 12-generation law. The 10 cM window mentioned later in the paper is for the wheat scan, not for this simulation, and was not used.
6. A named QTL is detected when the maximum likelihood ratio in $[0, 32]$, $(32, 60]$, or $(60, 100]$ exceeds the threshold. Effects are read at that argmax.

Seeds are `20101027` for this study. The outputs are `study1.json`, `study1_power.csv`, and `study1_effects.csv` under `simulation/out`.

For a complete recursive order, the Gaussian likelihood of the diagonal structural model equals the likelihood of the multivariate regression on the same exogenous regressors: the sum of the successive residual log variances is the log determinant of the residual covariance. The two joint likelihood ratios are therefore the same number. In the run, the maximum absolute difference of the two mean profiles is 7×10^{-14}. *(our own written proof, confirmed by our own simulation)* Mi et al. describe the two profiles as very close (PDF page 14). Their Table 1 powers still differ by 7, 8, and 2 replicates out of 100.

5 Replication results

Table 1 compares detection percentages. Our multi-trait powers keep the printed order, third QTL above the first two, and they sit 5 to 16 points below Table 1. The single-trait column for the first trait at the first QTL matches Table 1 exactly (6%). The single-trait column for the first trait at the third QTL does not: Table 1 prints 97% and the replication gives 13%.

Table 1: Detection percentages. "Mi" is Table 1 on PDF page 12. "Ours" is 100 replicates, seed `20101027`. SEM and LS coincide in our run.

QTL	SEM Mi	SEM ours	LS Mi	ST Y_1 Mi	ST Y_1 ours	ST overall ours
1	64	59	57	6	6	24
2	77	61	69	3	0	35
3	96	86	94	97	13	34

The algebraic total effects under $B = -M$ are, by columns of G,

$$(I + M)^{-1}G = \begin{pmatrix} -1/8 & 0 & 1/4 \\ 3/8 & 7/16 & 3/8 \\ 1/4 & 1/8 & 1/4 \end{pmatrix}.$$

(our own calculation) Table 2's parameter block matches this matrix on Y_2 and Y_3 for the first and third QTLs, and it does not match it on Y_1. The printed total effect of the third QTL on Y_1 is 0.75, against 0.25 from this first reading. Section 6 tests that disagreement directly.

Direct-effect means at the SEM argmax are close to G where the signal is large. For the third QTL they are 0.460 (0.175), 0.476 (0.179), and 0.242 (0.190), against the printed SEM estimates 0.479 (0.176), 0.527 (0.166), and 0.257 (0.205) and against $G = (1/2, 1/2, 1/4)$. *(our own simulation, set next to a cited source)* Position means are 22.65 (7.28), 41.26 (5.25), and 79.85 (3.76), against the printed 21.57 (2.83), 41.95 (2.58), and 78.03 (2.16). The biases are small. Our standard deviations are larger. The mean profile exceeds 14.86 only at 40 cM and on 75–82 cM. Figure 2 is described as one simulated population, with excursions on 16–28, 38–48, and 70–88 cM. An average profile is a different object, and ours is lower.

6 Which hypothesis closes which gap

Five changes were rerun from fixed seeds. The full grid is `simulation/out/gaps.json` and `gaps_table.csv`. Closure is $1 - |\hat{p} - p_{\text{printed}}|/|p_{\text{first}} - p_{\text{printed}}|$, using the first reading's 100-replicate powers as the denominator. A value of 1 means the printed number is met. A negative value means the estimate moved away from the table. *(our own simulation)*

Sign convention. $B = +M$ with unsigned G gives total effect 0.875 on Y_1 for QTL3, not 0.75. The glyph reading gives

$$(I - M)^{-1}G = \begin{pmatrix} 1/8 & 0 & 3/4 \\ 3/8 & -7/16 & 3/8 \\ 1/4 & -1/8 & 1/4 \end{pmatrix}.$$

(our own calculation) Every Table 2 parameter cell matches this arithmetic except the QTL2 row for Y_2, where the exact values are $-7/16 = -0.4375$ and $1/16 = 0.0625$ and the table prints -0.435 and 0.065. *(source discrepancy (our reading))* The first reading's total of 0.25 is what produced the single-trait power 13%. Under the glyph reading, 1000 replicates (seed `20101041`) give 86.2%

Table 2: Gap closure. SEM powers are the three QTLs. ST_3 is single-trait power for QTL3 on Y_1 (printed 97). ST_2 is single-trait power for QTL2 on Y_1 (printed 3). SDs are the three SEM position standard deviations. T_{31} is the algebraic total effect of QTL3 on Y_1 (printed 0.75).

Reading	SEM	ST_3	ST_2	position SD	T_{31}	ST_3 closed	T_{31} closed
Printed	64, 77, 96	97	3	2.83, 2.58, 2.16	0.75	—	—
First, $n = 100$	59, 61, 86	13	0	7.28, 5.25, 3.76	0.25	0	0
First, $n = 1000$	65, 63, 87	9	1	6.20, 5.32, 4.38	0.25	−0.05	0
$B = +M$, G unsigned	68, 57, 86	87	44	5.90, 5.33, 4.60	0.875	0.88	0.75
Glyph reading, $n = 100$	46, 45, 80	85	1	5.57, 5.93, 5.10	0.75	0.86	1
Glyph, $n = 1000$	52, 48, 87	86	1	5.15, 5.42, 3.47	0.75	0.87	1
Glyph, no cofactors	46, 65, 100	100	65	6.99, 9.12, 3.80	0.75	0.96	1
Glyph, wheat cofactors	34, 24, 89	95	1	7.33, 7.91, 3.77	0.75	0.98	1
Glyph, $\Psi = I$	80, 78, 98	97	2	4.86, 4.02, 2.79	0.75	1.00	1
Glyph, window ±5 cM	43, 41, 80	83	1	2.80, 2.76, 2.03	0.75	0.83	1
Glyph, $\alpha = 0.05$	77, 84, 98	95	14	5.57, 5.93, 5.10	0.75	0.98	1

(Clopper–Pearson interval 83.9–88.3). That closes most of the 97 versus 13 gap and does not meet 97.

Scan method. The simulation paragraph on PDF page 10 fits every marker except the flanking pair, and the methods section replaces the QTL genotype by its flanking-marker expectation (Haley and Knott, 1992). Dropping the cofactors, on the glyph reading, raises the single-trait QTL3 power to 100% and the SEM power for QTL3 to 100%, and it leaves the first SEM power at 46. The wheat rule on PDF page 11, a $P < 0.1$ screen with at most five cofactors and a 10 cM window, gives single-trait power 95% (interval 88.7–98.4, which contains 97) and SEM powers 34, 24, and 89. Nearest-marker regression gives the same pattern as dropping cofactors. None of these is the sentence printed for the simulation, and none of them brings all three SEM powers onto 64, 77, and 96.

Threshold and window. One hundred phenotype shuffles of a glyph-reading replicate have 95% quantile 14.74 for the maximum likelihood ratio, next to the printed 14.86. A permutation threshold at $\alpha = 0.05$ does not move the powers. The 500 permutations in the paper are in the wheat section. Replacing 14.86 and 10.60 by the $\alpha = 0.05$ quantiles on 4 and 2 degrees of freedom (9.49 and 5.99) raises the glyph-reading SEM powers to 77, 84, and 98 and the single-trait QTL3 power to 95. That threshold is not the printed one. Taking the peak within 5 cM of the true site, under the glyph reading, gives position standard deviations 2.80, 2.76, and 2.03, against 2.83, 2.58, and 2.16. The midpoint windows used in the first replication were our assumption. The paper does not name the window that assigns a peak to a named QTL.

Unprinted design. Eleven evenly spaced markers on a segment of 100 cM, with markers at the ends, are at $0, 10, \ldots, 100$. Shifting them to $5, 15, \ldots, 95$ does not close the SEM gap. Standardizing each trait to sample standard deviation 1 leaves the likelihood ratio essentially unchanged. Replacing Ψ by I does close the single-trait QTL3 gap (97%) and puts the second and third SEM powers on 78 and 98, while the first SEM power rises to 80 (interval 70.8–87.3, which does not contain 64). The simulation sentence says the residuals are drawn from matrix (8). $\Psi = I$ contradicts that sentence. It is a sensitivity, not the display.

Monte Carlo error. At 1000 replicates the first reading gives SEM powers 65.1, 62.5, and 86.6, with intervals 62.1–68.1, 59.4–65.5, and 84.3–88.7. The printed 64 lies in the first interval. The printed 77 and 96 do not. Single-trait power for QTL2 on Y_1 is 1.2% (interval 0.6–2.1) under that reading and 0.8% (0.3–1.6) under the glyph reading. The printed 3 is inside the $n = 100$ interval for zero detections (0–3.6) and outside the $n = 1000$ intervals. The discrepancy is three replicates in Table 1.

The most likely account is the glyph reading of equation (7), together with a peak window of about 5 cM that the paper does not state. That reading removes the 0.75 discrepancy and most of the single-trait QTL3 discrepancy, and the narrow window removes the position-SD discrepancy. It does not remove the SEM gap for QTL2 and QTL3. Those two percentages move onto the printed values only in runs that drop the printed cofactors or replace matrix (8). Both of those runs conflict with a sentence on PDF page 10.

7 Correlated residuals

The bow drops the edge into trait 0 from trait 2, sets that structural coefficient to zero, and frees the residual covariance $c = \rho\sqrt{1.6 \cdot 2.5}$. The other equation-(7) coefficients stay. The exogenous variables are the three QTL genotypes on the RIL map, not a marker scan. *(our assumption, matching* `recoverBow`*)* The sample is the centered second-moment matrix, $n = 150$, 400 replicates, seed `20101028`.

At the population covariance the bow plug-in recovers the parameter to machine precision. The diagonal plug-in does not. Its probability limit for the missing edge is 0.8ρ: zero at $\rho = 0$, 0.20 at $\rho = 0.25$, and 0.48 at $\rho = 0.6$. The QTL coefficient $G_{11} = 1/8$ has probability limit 0.075 at $\rho = 0.25$ and 0.005 at $\rho = 0.6$. The coefficient $G_{13} = 1/2$ moves to 0.45 and 0.38. *(our own simulation)* A Wald test that the missing edge is zero, using the diagonal delta-method standard error, rejects at rate 0.058 when $\rho = 0$ and at rate 0.728, 0.877, and 1 at $\rho = 0.25$, 0.30, and 0.60. The joint nine-degree Wald test of all QTL coefficients is only mildly inflated at $\rho = 0.25$ (rejection rate 0.065) and reaches 0.297 at $\rho = 0.6$. The damage shows up first as bias in two coefficients and as a spurious edge, not as an immediate explosion of the joint QTL test.

`recoverBow` is centered on the QTL and path coefficients. At $\rho = 0.25$ the Monte Carlo means of b_{12}, b_{23}, G_{11}, and G_{13} are -0.507, -0.498, 0.127, and 0.509. Coverage of the nominal 95% delta-method intervals, with the Gaussian Isserlis covariance plugged into the sample, is between 0.915 and 0.958 across the reported coefficients and the six correlations. The genotypes are ± 1, so Isserlis is an approximation. *(our assumption inside the standard error)* The coverage near 95% says the approximation is tolerable at $n = 150$ for these coefficients. Variance-component means sit a little low (1.54 against 1.6 at $\rho = 0$) because the likelihood uses divisor n; the probability limit is still the population value.

The same seeds were rerun with the glyph-reading coefficients: $b_{12} = +1/2$, $b_{23} = -1/2$, the edge b_{13} removed from $+1/4$, and the signed G. *(our own simulation)* At $\rho = 0.25$ the diagonal probability limit of the missing edge is again 0.20 and the Wald rejection rate is 0.728. `recoverBow` means are $b_{12} = 0.493$, $b_{23} = -0.498$, $G_{11} = -0.123$, and $G_{22} = -0.508$, with coverage between 0.920 and 0.955. The sign reading changes the parameter values and does not change the bias or the coverage.

8 Calibration to three phenotypic correlations

Campbell et al., Crop Science 43 (2003), Table 2, print phenotypic correlations for 95 recombinant inbred chromosome lines pooled across seven environments. There are no standard deviations and no environment or block adjustment. *(taken from a cited source, with the limits just stated)* The text layer of the copy we could read dropped minus signs. The paragraph above the table says that SPSM and kernels per square meter were negatively correlated with thousand-kernel weight. We therefore use $\mathrm{Corr}(\mathrm{TKWT}, \mathrm{SPSM}) = -0.44$. The cells $\mathrm{Corr}(\mathrm{TKWT}, \mathrm{KPS}) = 0.08$ and $\mathrm{Corr}(\mathrm{KPS}, \mathrm{SPSM}) = 0.59$ are the signs in that text layer; the prose does not restate them.

(our assumption for those two signs) GYLD is the fourth trait in the application and is not in equation (7). The three-trait order is Y_1 = TKWT, Y_2 = KPS, Y_3 = SPSM, which is the chain used in the indirect-effect arithmetic on PDF page 17 of Mi et al. *(our assumption)*

Treating the correlation matrix as a covariance, the diagonal recursive factorization is

$$B = \begin{pmatrix} 0 & 0.521 & -0.747 \\ 0 & 0 & 0.590 \\ 0 & 0 & 0 \end{pmatrix}, \qquad \Psi = \mathrm{diag}(0.629, 0.652, 1).$$

(our own calculation) The eigenvalues of the correlation are 0.224, 1.077, and 1.699, so this sign choice is positive definite. Naive Gaussian standard errors that pretend the matrix is a sample correlation of 95 i.i.d. lines are about 0.10 for the two coefficients in the first equation and 0.084 for 0.590. Campbell pooled seven environments, so those standard errors are not his sampling variances. The point fit does not depend on N.

The scan was rerun with this B and Ψ and with the signed G from the glyph reading of equation (7). Seed `20101029`, 100 replicates. Those QTL coefficients were not estimated from the correlations, and on this residual scale they are large: multi-trait power is 93%, 95%, and 99%. Single-trait power for QTL3 on Y_1 is 96%. That is a consequence of the scale mixture, not evidence that the 2003 correlations produce Mi's powers. *(our own simulation, under an assumption we flag)* The bow check at $\rho = 0.25$ (seed `20101030`) gives a diagonal probability limit 0.198 for the missing edge and a rejection rate 0.74. The unsigned-G run of the same seed had multi-trait powers 100%, 96%, and 100%. The sign change does not turn the calibration into a replication.

9 Limitations

The wheat fit is still blocked. It needs the 98 line identifiers, the ANOVA residuals of grain yield, thousand-kernel weight, kernels per spike, and spikes per square meter, the ± 1 QTL and cofactor codes, the divisor-n sample covariance, and the edge set as a matrix. Figure 5 draws arrows and does not print that matrix. Campbell's Table 2 does not substitute for any of those columns.

The Drton–Foygel–Sullivant criterion is not a theorem of this repository. The delta method is not a Lean central limit theorem. The glyph reading of equation (7) meets Table 2's parameter block up to the two rounded QTL2 entries. It leaves the SEM powers for QTL2 and QTL3, and about eleven points of the single-trait QTL3 power, short of Table 1. Those leftovers are not Monte Carlo error at 1000 replicates. Twelve generations of selfing, Haldane's map, the marker placement, and the detection windows are our assumptions. The 2021 Hidalgo-Contreras wheat SEM is a different experiment.

10 Reproducibility

From `formalization/correlated-residuals/simulation`,

```
python3 studies.py --reps 100 --bow-reps 400 --studies 1,2,3
```

writes `study1.json`, `study2.json`, and `study3.json`. The CSV tables are built from those files.

- Simulation commit: `5cd3c9f`.
- Full hash: `5cd3c9f6d2f6e38f25e32146d74d55f3778436e1`.
- Branch: `cursor/l-f102-correlated-residuals-4c01`.

- Lean 4.22.0 and Mathlib v4.22.0, built with `lake build`.
- Theorem links are pinned to `05d69ea141adf3d5ee936d3b48061bde241250dc`.
- This PDF is `pdflatex paper.tex`. The ledger is `SOURCES.md`.
- The gap grid is `python3 gaps.py` in the simulation directory. Seeds are `20101027` through `20101042`.

A Central Lean proofs

The diagonal model reads the structural coefficients from the Schur complement $\Omega = \Sigma_{yy} - \Sigma_{yx}\Sigma_{xx}^{-1}\Sigma_{yx}^{\mathsf{T}}$, then recovers $\Gamma = (I - B)\Sigma_{yx}\Sigma_{xx}^{-1}$ and $\Phi = \Sigma_{xx}$. The bow-free model drops the edge into the first trait from the third and reads the residual covariance c from the same complement. The collision proofs exhibit, for an arbitrary exogenous coefficient, a second parameter point with the same joint blocks. Consistency is continuous mapping of those left inverses, composed with the strong law for the sample Gram matrix. The listings below are the theorem bodies. Their lemmas live in the linked files.

`recover_sigma`, `Diagonal.lean`, lines 246–279.

```
theorem recover_sigma (θ : Params) (h : OnDomain θ) : recover (sigma θ) = θ := by
  obtain ⟨hv1, hv2, hv3, hφ⟩ := h
  have _hpsi : (psi θ).PosDef := by
    rw [psi, posDef_diagonal_iff]
    intro i
    fin_cases i
    · simpa using hv1
    · simpa using hv2
    · simpa using hv3
  have hread : readDiag (omega θ) = (θ.b12, θ.b13, θ.b23, θ.v1, θ.v2, θ.v3) :=
    readDiag_omega θ hv2 hv3
  have hschur := schur_omega θ hφ
  simp only [recover, hschur, hread]
  have hB : (1 -
      ![0, θ.b12, θ.b13;
         0, 0, θ.b23;
         0, 0, 0]) = subB θ := by
    simpa [Bmat] using (subB_eq_one_sub θ).symm
  have hdet : IsUnit θ.phi.det := (isUnit_iff_isUnit_det θ.phi).1 hφ.isUnit
  apply Params.ext
  · rfl
  · rfl
  · rfl
  · rfl
  · rfl
  · rfl
  · simp only [sigma]
    rw [hB]
    calc
      subB θ * (Lmat θ * θ.gamma * θ.phi) * θ.phi⁻¹
          = (subB θ * Lmat θ) * θ.gamma * (θ.phi * θ.phi⁻¹) := by simp only [mul_assoc]
      _ = (1 : M3) * θ.gamma * (1 : M3) := by rw [subB_mul_L, mul_nonsing_inv _ hdet]
      _ = θ.gamma := by simp
  · rfl
```

`recover_bow`, `BowFree.lean`, lines 187–213.

```
theorem recover_bow (θ : Bow) (h : OnBow θ) : recoverBow θ.sigma = θ := by
  obtain ⟨hN, hφ⟩ := h
  have hv2 := Bow.v2_pos θ hN
  have hv3 := Bow.v3_pos θ hN
  have hread := readBow_omega θ hv2 hv3
  have hschur : θ.sigma.yy - θ.sigma.yx * θ.sigma.xx⁻¹ * θ.sigma.yxᵀ = θ.omega := by
    simpa [Bow.sigma, Bow.omega] using
      schur_noise (L := θ.L) (G := θ.gamma) (Φ := θ.phi) (N := θ.noise) hφ
  simp only [recoverBow, hschur, hread]
  have hB : (1 - !![0, θ.b12, 0; 0, 0, θ.b23; 0, 0, 0]) = θ.subB := by
    simpa [Bow.B] using (Bow.subB_eq θ).symm
  have hdet : IsUnit θ.phi.det := (isUnit_iff_isUnit_det θ.phi).1 hφ.isUnit
  apply Bow.ext
  · rfl
  · rfl
  · rfl
  · rfl
  · rfl
  · rfl
  · simp only [Bow.sigma]
    rw [hB]
    calc
      θ.subB * (θ.L * θ.gamma * θ.phi) * θ.phi⁻¹
          = (θ.subB * θ.L) * θ.gamma * (θ.phi * θ.phi⁻¹) := by simp only [mul_assoc]
      _ = (1 : M3) * θ.gamma * (1 : M3) := by rw [Bow.subB_mul_L, mul_nonsing_inv _ hdet]
      _ = θ.gamma := by simp
  · rfl
```

`bow12_same_law` in `Collision.lean`, lines 114–128. The siblings `bow13_same_law` and `bow23_same_law` repeat the argument on the other two pairs, lines 130–160.

```
theorem bow12_same_law (c : ℝ) (G Φ : M3) :
    bow12L c * (bow12G c * G) * Φ = G * Φ ∧
      bow12L c * ((bow12G c * G) * Φ * (bow12G c * G)ᵀ + bow12G c * (bow12G c)ᵀ) * (bow12L c)ᵀ =
        G * Φ * Gᵀ + 1 := by
  have hG : bow12L c * (bow12G c * G) = (1 : M3) * G := by
    rw [← mul_assoc, bow12_resolve c, one_mul]
  have hN : bow12L c * (bow12G c * (bow12G c)ᵀ) * (bow12L c)ᵀ = (1 : M3) :=
    resolved_noise (bow12_resolve c)
  have hbaseN : (1 : M3) * (1 : M3) * (1 : M3)ᵀ = (1 : M3) := by simp
  obtain ⟨hyx, hyy⟩ := exogenous_lift (L := bow12L c) (G := bow12G c * G)
    (L' := (1 : M3)) (G' := G) (N := bow12G c * (bow12G c)ᵀ) (N' := (1 : M3)) (Φ := Φ)
    hG (hN.trans hbaseN.symm)
  constructor
  · simpa using hyx
  · simpa using hyy
```

`unitUpper_factor_unique`, `Triangular.lean`, lines 68–135.

```
theorem unitUpper_factor_unique {n : ℕ}
    {L M : Matrix (Fin n) (Fin n) ℝ} {d e : Fin n → ℝ}
    (hL : UnitUpper L) (hM : UnitUpper M)
    (hd : ∀ i, 0 < d i) (he : ∀ i, 0 < e i)
    (hΩ : L * diagonal d * Lᵀ = M * diagonal e * Mᵀ) :
    L = M ∧ d = e := by
  induction n with
  | zero =>
```

```
  constructor
  · ext i j
    exact i.elim0
  · ext i
    exact i.elim0
| succ n ih =>
  have hde_last : d (Fin.last n) = e (Fin.last n) := by
    have hLlast := gram_last (d := d) hL (Fin.last n)
    have hMlast := gram_last (d := e) hM (Fin.last n)
    rw [hL.2] at hLlast
    rw [hM.2] at hMlast
    have hentry := congr_fun (congr_fun hΩ (Fin.last n)) (Fin.last n)
    rw [hLlast, hMlast] at hentry
    simpa using hentry
  have hcol : ∀ i, L i (Fin.last n) = M i (Fin.last n) := by
    intro i
    have hLi := gram_last (d := d) hL i
    have hMi := gram_last (d := e) hM i
    have hentry := congr_fun (congr_fun hΩ i) (Fin.last n)
    rw [hLi, hMi, hde_last] at hentry
    exact mul_right_cancel₀ (he (Fin.last n)).ne' hentry
  have hlead :
      lead L * diagonal (fun k : Fin n => d k.castSucc) * (lead L)ᵀ =
        lead M * diagonal (fun k : Fin n => e k.castSucc) * (lead M)ᵀ := by
    ext i j
    have hLij := lead_gram (L := L) (d := d) i j
    have hMij := lead_gram (L := M) (d := e) i j
    have hentry := congr_fun (congr_fun hΩ i.castSucc) j.castSucc
    rw [hLij, hMij] at hentry
    have hr :
        L i.castSucc (Fin.last n) * d (Fin.last n) * L j.castSucc (Fin.last n) =
          M i.castSucc (Fin.last n) * e (Fin.last n) * M j.castSucc (Fin.last n) := by
      rw [hcol, hcol, hde_last]
    rw [hr] at hentry
    exact add_right_cancel hentry
  have hrec :=
    ih (lead_unitUpper hL) (lead_unitUpper hM) (fun i => hd i.castSucc)
      (fun i => he i.castSucc) hlead
  refine ⟨?_, ?_⟩
  · ext i j
    cases i using Fin.lastCases with
    | last =>
      cases j using Fin.lastCases with
      | last =>
        rw [hL.2, hM.2]
      | cast j =>
        rw [hL.1 _ _ j.castSucc_lt_last, hM.1 _ _ j.castSucc_lt_last]
    | cast i =>
      cases j using Fin.lastCases with
      | last =>
        exact hcol i.castSucc
      | cast j =>
        have hlead_ij := congr_fun (congr_fun hrec.1 i) j
        simpa [lead, submatrix_apply] using hlead_ij
  · ext i
    cases i using Fin.lastCases with
    | last => exact hde_last
    | cast i =>
      have := congr_fun hrec.2 i
```

```
        simpa using this
```

sigmaRec_injective, Triangular.lean, lines 298–317.

```
theorem sigmaRec_injective {n p : ℕ} {θ θ' : RecModel n p}
    (hθ : OnRec θ) (hθ' : OnRec θ') (h : sigmaRec θ = sigmaRec θ') : θ = θ' := by
  have hθ0 := hθ
  have hθ'0 := hθ'
  obtain ⟨hB, hd, hΦ⟩ := hθ
  obtain ⟨hC, he, hΦ'⟩ := hθ'
  have hgram :
      (1 - θ.B)⁻¹ * diagonal θ.d * ((1 - θ.B)⁻¹)ᵀ =
        (1 - θ'.B)⁻¹ * diagonal θ'.d * ((1 - θ'.B)⁻¹)ᵀ := by
    have hθs := schur_sigmaRec θ hΦ
    have hθs' := schur_sigmaRec θ' hΦ'
    rw [h] at hθs
    exact hθs.symm.trans hθs'
  obtain ⟨hBeq, hdeq⟩ := recursive_residual_unique hB hC hd he hgram
  have hγ : θ.gamma = θ'.gamma := by
    have hg := (recover_exogenous θ hθ0).2
    have hg' := (recover_exogenous θ' hθ'0).2
    rw [hg, hg', h, hBeq]
  have hφ : θ.phi = θ'.phi := by simpa [sigmaRec] using congrArg Joint.xx h
  exact RecModel.ext hBeq hdeq hγ hφ
```

plug_in_consistent_ae, Law.lean, lines 174–199.

```
theorem plug_in_consistent_ae {Ω : Type*} [MeasurableSpace Ω] {μ : Measure Ω}
    {Z : ℕ → Ω → Idx → ℝ} {θ : Params} (hθ : OnDomain θ)
    (hindep : Pairwise fun i j => IndepFun (Z i) (Z j) μ)
    (hident : ∀ i, IdentDistrib (Z i) (Z 0) μ μ)
    (hint : ∀ a b : Idx, Integrable (fun ω => Z 0 ω a * Z 0 ω b) μ)
    (hyy : ∀ i j : Fin 3,
      ∫ ω, Z 0 ω (.inl i) * Z 0 ω (.inl j) ∂μ = (sigma θ).yy i j)
    (hyx : ∀ i j : Fin 3,
      ∫ ω, Z 0 ω (.inl i) * Z 0 ω (.inr j) ∂μ = (sigma θ).yx i j)
    (hxx : ∀ i j : Fin 3,
      ∫ ω, Z 0 ω (.inr i) * Z 0 ω (.inr j) ∂μ = (sigma θ).xx i j) :
    ∀ᵐ ω ∂μ, Tendsto (fun n =>
        ((recover (sampleCov Z n ω)).b12, (recover (sampleCov Z n ω)).b13,
          (recover (sampleCov Z n ω)).b23, (recover (sampleCov Z n ω)).v1,
          (recover (sampleCov Z n ω)).v2, (recover (sampleCov Z n ω)).v3,
          (recover (sampleCov Z n ω)).gamma, (recover (sampleCov Z n ω)).phi))
      atTop
      (nhds (θ.b12, θ.b13, θ.b23, θ.v1, θ.v2, θ.v3, θ.gamma, θ.phi)) := by
  filter_upwards [strong_gram Z hindep hident hint] with ω hω
  have hyy' : Tendsto (fun n => (sampleCov Z n ω).yy) atTop (nhds (sigma θ).yy) := by
    simpa [sampleCov, blockYY_pop hyy] using tendsto_blockYY hω
  have hyx' : Tendsto (fun n => (sampleCov Z n ω).yx) atTop (nhds (sigma θ).yx) := by
    simpa [sampleCov, blockYX_pop hyx] using tendsto_blockYX hω
  have hxx' : Tendsto (fun n => (sampleCov Z n ω).xx) atTop (nhds (sigma θ).xx) := by
    simpa [sampleCov, blockXX_pop hxx] using tendsto_blockXX hω
  exact plug_in_consistent hθ hyy' hyx' hxx'
```

plug_in_bow_consistent_ae, BowConsistency.lean, lines 270–304.

```
theorem plug_in_bow_consistent_ae {Ω : Type*} [MeasurableSpace Ω] {μ : Measure Ω}
    {Z : ℕ → Ω → Idx → ℝ} {θ : Bow} (hθ : OnBow θ)
```

```
  (hindep : Pairwise fun i j => IndepFun (Z i) (Z j) μ)
  (hident : ∀ i, IdentDistrib (Z i) (Z 0) μ μ)
  (hint : ∀ a b : Idx, Integrable (fun ω => Z 0 ω a * Z 0 ω b) μ)
  (hyy : ∀ i j : Fin 3,
    ∫ ω, Z 0 ω (.inl i) * Z 0 ω (.inl j) ∂μ = θ.sigma.yy i j)
  (hyx : ∀ i j : Fin 3,
    ∫ ω, Z 0 ω (.inl i) * Z 0 ω (.inr j) ∂μ = θ.sigma.yx i j)
  (hxx : ∀ i j : Fin 3,
    ∫ ω, Z 0 ω (.inr i) * Z 0 ω (.inr j) ∂μ = θ.sigma.xx i j) :
  ∀ᵐ ω ∂μ, Tendsto (fun n =>
      ((recoverBow (sampleCov Z n ω)).b12, (recoverBow (sampleCov Z n ω)).b23,
        (recoverBow (sampleCov Z n ω)).v1, (recoverBow (sampleCov Z n ω)).v2,
        (recoverBow (sampleCov Z n ω)).v3, (recoverBow (sampleCov Z n ω)).c,
        (recoverBow (sampleCov Z n ω)).gamma, (recoverBow (sampleCov Z n ω)).phi))
    atTop
    (nhds (θ.b12, θ.b23, θ.v1, θ.v2, θ.v3, θ.c, θ.gamma, θ.phi)) := by
filter_upwards [strong_gram Z hindep hident hint] with ω hω
have hyy' : Tendsto (fun n => (sampleCov Z n ω).yy) atTop (nhds θ.sigma.yy) := by
  have hpop : blockYY (popGram μ (Z 0)) = θ.sigma.yy := by
    ext i j
    simpa [blockYY, popGram] using hyy i j
  simpa [sampleCov, hpop] using tendsto_blockYY hω
have hyx' : Tendsto (fun n => (sampleCov Z n ω).yx) atTop (nhds θ.sigma.yx) := by
  have hpop : blockYX (popGram μ (Z 0)) = θ.sigma.yx := by
    ext i j
    simpa [blockYX, popGram] using hyx i j
  simpa [sampleCov, hpop] using tendsto_blockYX hω
have hxx' : Tendsto (fun n => (sampleCov Z n ω).xx) atTop (nhds θ.sigma.xx) := by
  have hpop : blockXX (popGram μ (Z 0)) = θ.sigma.xx := by
    ext i j
    simpa [blockXX, popGram] using hxx i j
  simpa [sampleCov, hpop] using tendsto_blockXX hω
exact plug_in_bow_consistent hθ hyy' hyx' hxx'
```

References

[1] C. Brito and J. Pearl. A new identification condition for recursive models with correlated errors. *Structural Equation Modeling* 9 (2002), 459–474.

[2] B. T. Campbell, P. S. Baenziger, K. S. Gill, K. M. Eskridge, H. Budak, M. Erayman, and Y. Yen. Identification of QTLs and environmental interactions associated with agronomic traits on chromosome 3A of wheat. *Crop Science* 43 (2003), 1493–1505.

[3] M. Drton, R. Foygel, and S. Sullivant. Global identifiability of linear structural equation models. *Annals of Statistics* 39 (2011), 865–886.

[4] R. Foygel, J. Draisma, and M. Drton. Half-trek criterion for generic identifiability of linear structural equation models. *Annals of Statistics* 40 (2012), 1682–1713.

[5] J. V. Hidalgo-Contreras, J. Salinas-Ruiz, K. M. Eskridge, and S. P. Baenziger. Incorporating molecular markers and causal structure among traits using a Smith–Hazel index and structural equation models. *Agronomy* 11 (2021), 1953.

[6] R. Li, S.-W. Tsaih, K. Shockley, I. M. Stylianou, J. Wergedal, B. Paigen, and G. A. Churchill. Structural model analysis of multiple quantitative traits. *PLoS Genetics* 2 (2006), e114.

[7] X. Mi, K. Eskridge, D. Wang, P. S. Baenziger, B. T. Campbell, K. S. Gill, I. Dweikat, and J. Bovaird. Regression-based multi-trait QTL mapping using a structural equation model. *Statistical Applications in Genetics and Molecular Biology* 9 (2010), article 38.

[8] B. D. Valente, G. J. M. Rosa, G. de los Campos, D. Gianola, and M. A. Silva. Searching for Recursive Causal Structures in Multivariate Quantitative Genetics Mixed Models. *Genetics* 185 (2010), 633–644.